PRISM

An AI assistant for in-line spectroscopy solution design:
from a customer enquiry to a grounded first technical assessment.

Rules decide, the LLM writes | **AFS evidence with verbatim quotes** | **Asks before it guesses** | **Every sentence verified**

APPROACH

A sales engineer's first assessment: grounded and honest

The task

- Read a free-text enquiry: process, sample, targets, constraints.
- Ask what is missing; recommend method, spectral range and probe concept.
- Justify with the AFS materials; risks; first feasibility study.

Why not a plain chatbot

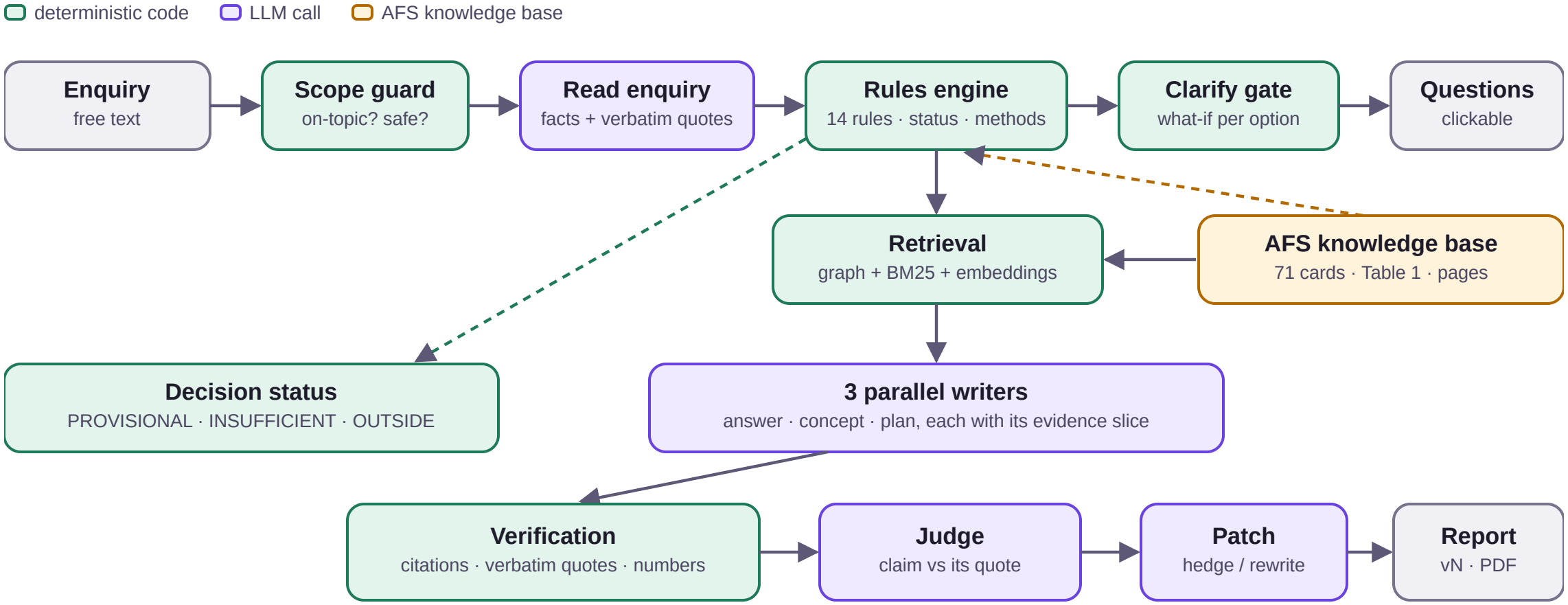
- Invents specifications and performance figures.
- Fills in missing client facts; quotes lab studies as proof.
- Sounds confident where "not enough information" is right.

PRISM's principles

- **Rules decide, the LLM writes.** Deterministic engine owns status, candidates, blocked methods.
- **AFS materials first.** AFS claims cite a card whose quote is verbatim on the page.
- **Typed statements:** client fact · AFS · external · assumption · proposal · unknown.
- **Ask before guessing:** questions ranked by how much the answer changes the recommendation.
- **Verify every sentence** before the scientist sees it.

SYSTEM ARCHITECTURE

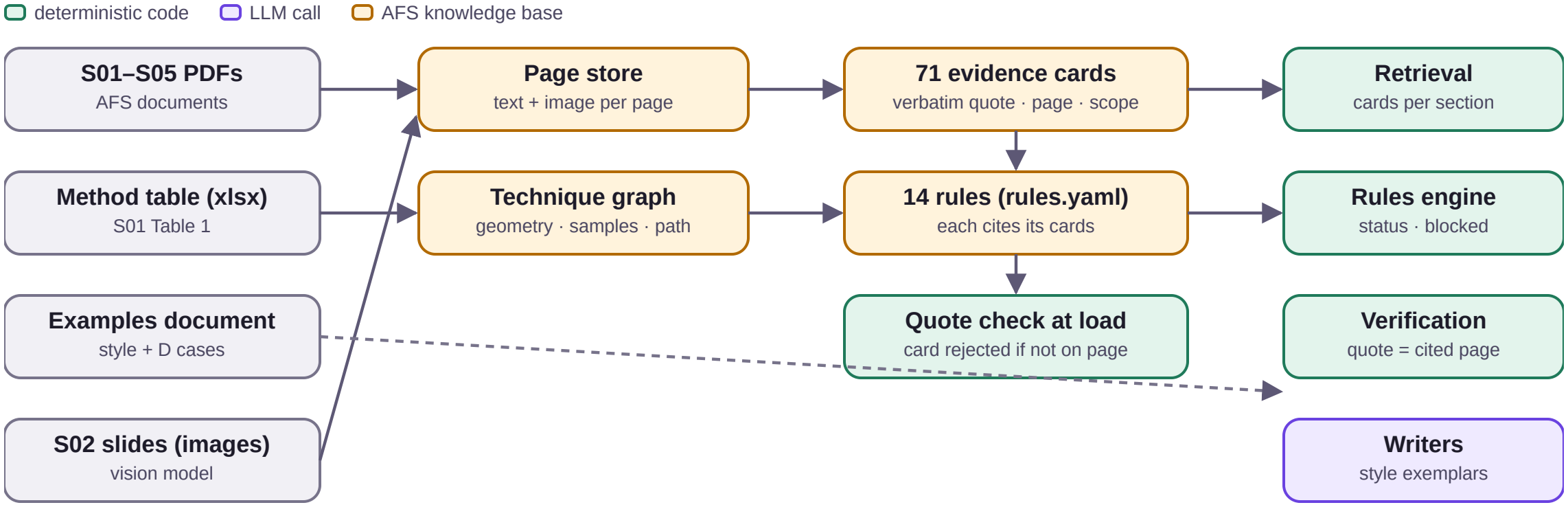
Neuro-symbolic pipeline



The LLM never decides the status: it reads the enquiry, writes the sections and judges claims. Everything that decides is code.

USE OF AFS MATERIALS

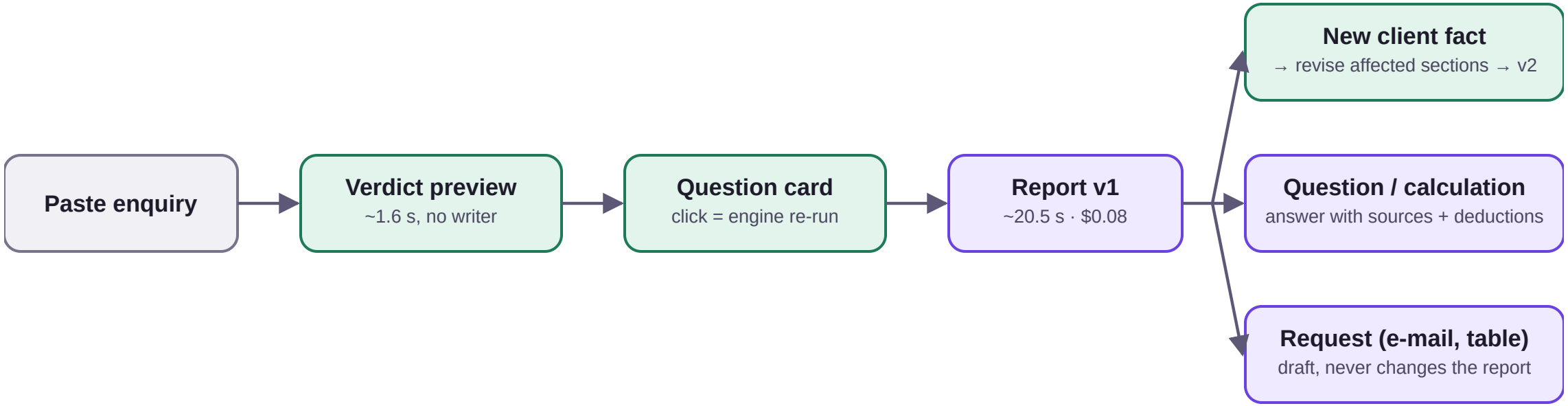
How the knowledge base is built and used



No case-specific hard-coding: the H01–H06 file is read only by the answer script; an n-gram check confirms no prompt, rule or ontology shares wording with it.

THE ASSISTANT

From enquiry to living report



Clarifying questions

Ranked by a what-if over the engine; each option shows what it changes. Clicks cost no LLM call.

Living report

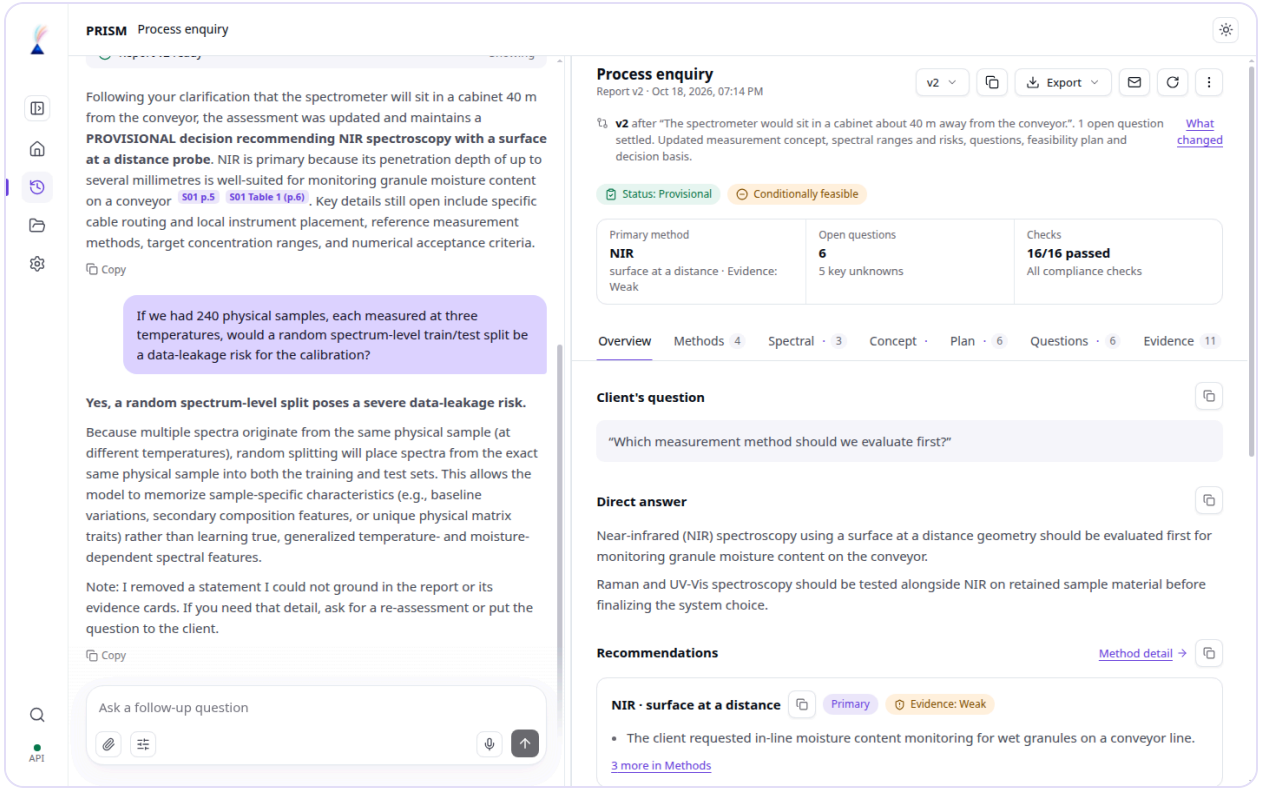
A new fact revises only the affected sections: v2 with "What changed". Major changes ask first.

Grounded follow-ups

AFS evidence, client facts, deductions and general knowledge are labelled, not refused.

THE PROTOTYPE

Built for scientists



- **Report tabs:** overview, methods, spectral, concept, plan, questions, evidence, checks.
- **Every citation** opens its AFS page, highlighted.
- **Exports:** A4 PDF, Markdown, JSON of the version on screen.
- **Own sources:** upload files and folders, add links; cited separately from AFS.
- **MCP integrations** with per-tool permissions and approvals.
- Voice input, dark mode, copy buttons.

EXAMPLE RECOMMENDATION · H04

"Can the assistant approve this concept?" Not yet.

Enquiry: water content in a petroleum line using MIR ATR; analyzer in another building; reuse a silica fiber from a VIS/NIR setup; link length and conditions not supplied.

INSUFFICIENT INFORMATION NOT YET

> There is not enough information to approve the MIR ATR concept as proposed.

CONTRADICTED reuse of the silica fiber:

> VIS–NIR silica transmits about 350–2400 nm, which does not overlap the MIR range of 4000–400 cm^{-1} (2.5–25 µm).

Concept

- MIR ATR probe (conditional) and screening of candidate methods on retained samples.
- Fiber: polycrystalline (AgHal) IR fiber or hollow glass waveguide; long link to be confirmed with AFS.

First questions to the client

- What does the liquid look like at the measurement point?
- Cable route and length to the other building?
- Research, pilot or production use?

Two dairy streams, two probe geometries, one comparison study

Recommendation

- **Clear whey:** NIR transmission / transflection probe (primary).
- **Opaque concentrate:** NIR diffuse reflection probe (primary).
- NIR 780–2500 nm; VIS-NIR silica fiber; link length to be confirmed.

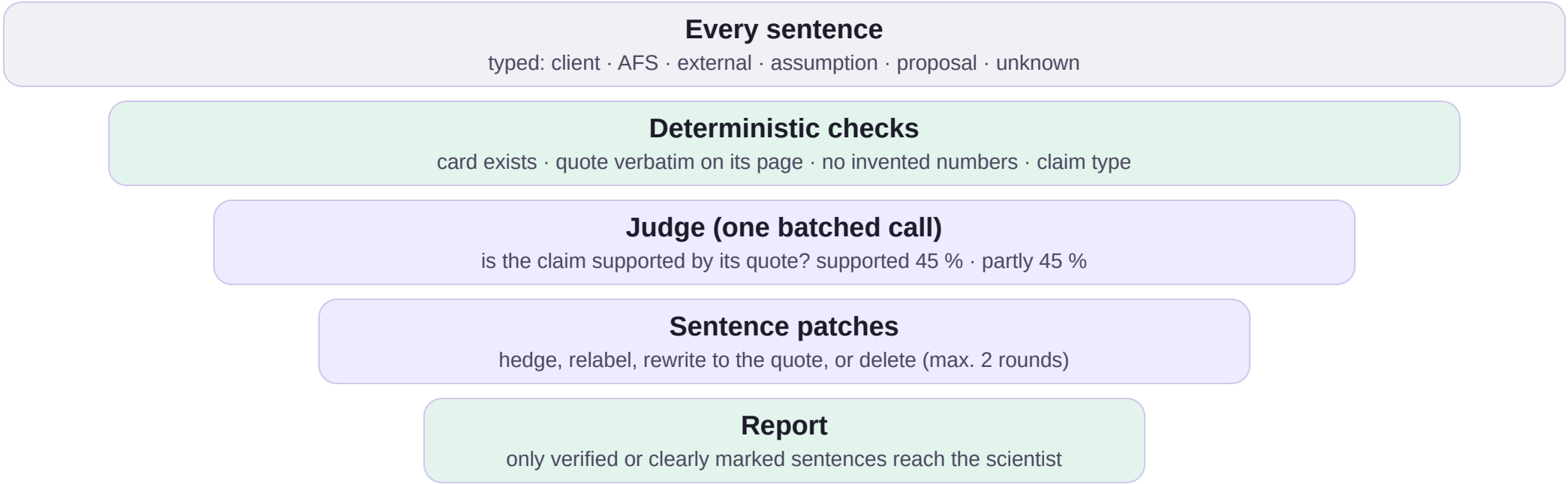
> Compare the specified NIR probe geometries on the same representative samples from each stream before selecting the installation.

First feasibility study (≤ 6 steps)

1. Clarify ranges, acceptance criteria and installation constraints.
2. Collect representative samples across agreed compositions and physical states.
3. Obtain time-aligned laboratory references.
4. Set up interfaces; compare spectra and models per stream.
5. Validate on independent production material.
6. Go / no-go against pre-agreed criteria.

TRUST

Verification: what reaches the scientist



Partly supported sentences are hedged or rewritten to the quote; unsupported ones are removed. The report states structural checks and evidence support separately.

FINAL EVALUATION SCENARIOS

H01–H06: generated by PRISM, replayable

CASE	TOPIC	STATUS	KEY POINT
H01	New formulation after an NIR pilot	PROVISIONAL	Test the old calibration on the new blend first; that adding Raman keeps it valid is unverified
H02	Two dairy streams	PROVISIONAL	Transmission for clear whey, reflection for the opaque concentrate; compare on both
H03	Fluorescence for glucose and biomass	PROVISIONAL	Fluorescence not recommended without a fluorescent target; NIR / Raman / UV-Vis to compare
H04	Silica link for MIR ATR	INSUFFICIENT	Silica fiber reuse contradicted (0.35–2.4 µm vs 2.5–25 µm); not approved
H05	Undefined off-spec liquid	INSUFFICIENT	No exact configuration yet; screening experiment and deciding questions first
H06	Research sensor for surgery	OUTSIDE SCOPE	In-vitro study is not proof of clinical readiness; research-grade feasibility only

All six pass every structural check. **make verify-outputs** replays them from the committed LLM cache and matches byte for byte (also in CI).

EVIDENCE IT WORKS

Evaluation: gold development cases and new follow-ups

8 / 8
decision status as expected (D01–D08)

0
invented numbers

100 %
citations exist · quotes verbatim · client facts kept

12 / 12
new follow-up questions handled correctly

Gold agreement (live, OpenAI)

- Method family 7/7 · premise verdicts 6/7 · must-cite pages 81 %.
- Traps handled: missing information, contradicted premises, fiber windows, evidence scope, multi-stream.

Generality test (new domain: polymer melt)

- Unit conversion → IR fiber · hypothetical · new fact → v2 · data-split advice · XRF (partly outside sources) · contradicted premise · table · e-mail.
- Prompt injection and off-topic refused.

COST AND TIME PER QUERY

Fast enough to use in a customer call



QUERY	TIME	MODEL CALLS	COST
Verdict preview + questions	~1.6 s	1	< $0.01
Clicking an answer option	< 0.1 s	0	$0
Full report (standard)	~20.5 s	7	$0.08
Follow-up turn (average)	4–30 s	1–8	~$0.09

- Measured on OpenAI: 18,799 input / 4,010 output tokens per report (GPT-5.4 writers, 5.4-mini reader and judge, 5.4-nano patcher).
- Cost = tokens × OpenAI list price (mid-2026). Deep mode uses gpt-6.1-sol, whose price is not published.
- Every call is cached: replaying a report costs nothing.

HOW IT WAS BUILT

From the AFS materials to a working assistant

1 · Knowledge

ingest S01–S05, Table 1 graph, 71 cards with page-checked quotes

2 · Rules

14 deterministic rules: geometry, fiber windows, premises, scope

3 · Pipeline

extract → rules → retrieve → write → verify → patch

4 · Assistant

clarify gate, living report, tools, approvals, exports

5 · Evaluation

gold D cases, synthetic traps, live follow-ups, replayable H answers

Test-driven

~590 backend and ~270 frontend tests, all offline with fake models; CI on every push.

Provider-agnostic

Roles map to models per profile: OpenAI, Gemini or a custom endpoint.

Measured

Live gold evaluation, a 12-question generality test and replayable official answers.

What PRISM does not do yet

Limitations

- **Coverage:** 71 curated cards; 129 drafted cards await expert review; S05/S06 are business context only.
- **Support strength:** under half of judged claims are fully supported by their quote; the rest are hedged or marked.
- **Premise verdicts** can still differ from an expert's (1 of 7 on the development cases); the evaluation set is small (8 gold cases).
- No measured spectra, no OCR; prototype without user accounts; enquiry text goes to the chosen LLM provider.

Next steps

- Expert review of drafted cards; page-level AFS retrieval.
- Calibrate the judge on expert labels (tooling in place).
- Attach measured spectra from the AFS lab to the plan.
- Accounts, roles and an on-premise model for NDA enquiries.